

Master’s Thesis Proposal

Harvard Master’s in Data Science

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1. Title

Attention-Guided Router Steering: A Feature-Learning Lens on Behavioral Intervention in Sparse Mixture-of-Experts Language Models.

2. Advisor

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Pending administrative confirmation with Daniel Weinstock as to whether the Harvard MS in Data Science program requires an internal SEAS sponsor when the primary advisor is an MIT faculty member. If so, a Harvard co-advisor will be added before the proposal is countersigned.

3. Abstract

Sparse mixture-of-experts (MoE) language models such as Mixtral 8x7B route every token to a small subset of feed-forward experts via a learned router. Recent work (STEERMOE) showed that the router itself is a viable intervention site: by comparing routing on contrastive prompt pairs we can identify experts whose activation rate differs in the presence of a target concept and bias routing toward them at decode time. In a faithful re-implementation I find that this recipe transfers *weakly* to Mixtral on a question-only concept-transfer test, and that the underlying signal is diffuse: the top twenty (layer, expert) cells out of $32 \times 8 = 256$ account for less than ten percent of the total $|\Delta_{\ell,e}|$ mass. This thesis proposes **Attention-Guided Router Steering (AGRS)**, a drop-in replacement for STEERMOE in which the per-token contribution to the readout is weighted by the attention mass that token sends toward a concept-bearing prefix span. AGRS reduces to STEERMOE when the attention weights are uniform, which makes the contribution isolatable from confounding implementation choices. I evaluate AGRS against STEERMOE and three hidden-state steering baselines across four MoE checkpoints and four concept families, with a pre-registered Phase-3 decision gate ($\Delta\text{CTR} \geq +0.10$) before the more expensive generalization runs. The mechanistic chapter uses the Average Gradient Outer Product (AGOP) framework developed in my advisor’s recent work to ask whether AGRS’s expert selections align with concept-distinctive feature directions in a way STEERMOE’s do not. Deliverables are a NeurIPS-track method paper, an open-source

replication artifact (Apptainer image, pinned model SHAs, full judge transcripts), and a written monograph integrating the empirical headline with the feature-learning lens.

4. Motivation and Background

4.1 Who I am

I am a candidate in Harvard’s Master’s in Data Science program. My prior research experience is a project on hyperbolic prototype embeddings for hierarchical artistic style on WikiArt,¹ which trained the same prototype classifier in matched-dimension Euclidean and Poincaré-ball output spaces and reported a robust local-retrieval advantage for the hyperbolic geometry across 150 seed-replicated runs. That project taught me how to design a tightly scoped representation-learning study, run an honest mixed-results comparison across 18 paired configurations, and ship a reproducible end-to-end artifact — the skills directly transferable to the thesis proposed here.

4.2 Why this project

For the past several months I have been running empirical experiments on STEERMoe-style routing-bias interventions on Mixtral 8x7B-Instruct on the MIT ORCD cluster. A working draft of the conference paper that came out of those experiments is checked into the project repository.² That draft documents a specific limitation of the upstream method, which is the empirical hook the thesis turns on.

Three converging observations made this the project I want to spend my thesis year on:

- Steering is the cleanest empirical wedge into the question of *what features a model has learned*. If we can move a generation toward a concept by adding a bias to a small set of scalar parameters at inference time, then those parameters carry the concept in a behaviorally-meaningful sense; the steering experiment is a causal probe of representation.
- Sparse MoE language models are an unusually clean substrate for that probe. Every layer’s routing decision is a discrete, inspectable choice over a small expert pool, unlike the continuous residual stream of a dense transformer. The routing layer thus gives steering research a structurally different intervention site than any of the existing dense-transformer methods.
- Existing steering methods read off the contrastive signal using fixed token positions or unweighted averages over a span. Attention-guided readouts have been demonstrated for hidden-state interventions on dense transformers but never for router-logit interventions on sparse MoE models. The gap is identifiable, and closing it is the natural next contribution.

4.3 Preliminary results

The empirical hook for the thesis is concrete. Across 25 question-only test cases (5 fear concepts $\times$ 5 evaluation prompts) on Mixtral 8x7B-Instruct, swept across eight (α, k_+, k_-) configurations of STEERMoe, three things hold consistently:

¹Repository and preprint: github.com/pgrindehollevik-harvard/hyperbolic.

²Working draft: `paper/docs/paper/main.pdf` in github.com/pgrindehollevik-harvard/MoE_attention_guided_steering (paper-submission branch).

1. The *concept-token hit rate* —the fraction of cases whose response contains any token of the target concept name — moves by at most 0.04 between baseline and STEERMOE. Most configurations show zero shift.
2. Response length contracts monotonically with the steering coefficient α , from -2.16 words at $\alpha = 1$ to -5.52 words at $\alpha = 0.01$ with a large k_+ , indicating the intervention has a measurable but defocusing effect.
3. The per-(layer, expert) risk-difference signal that drives the STEERMOE plan is *diffuse* on Mixtral. The top twenty cells out of the 256 total capture less than 10% of the total $|\Delta_{\ell,e}|$ mass — there is no small subset of experts that explains most of the prefix-conditioned vs. unprefixed routing gap.

The three observations together motivate AGRS: if the readout is diffuse and the intervention barely moves the model, then the readout is the bottleneck, and reweighting it by attention to the concept span is the lightest-touch method change that could plausibly fix the diffusion.

4.4 How this project fits my advisor’s research

Adit’s recent work — *Mechanism for Feature Learning in Neural Networks* (Science, 2024), *Mechanism of Feature Learning in Convolutional Neural Networks* (arXiv:2309.00570), and *Linear Recursive Feature Machines Provably Recover Low-rank Matrices* (arXiv:2401.04553) — introduces the **Average Gradient Outer Product (AGOP)** as a mathematical mechanism by which neural networks acquire features. The AGOP matrix $M = \mathbb{E}_x [\nabla_x f(x) \nabla_x f(x)^\top]$ encodes *which input directions a model treats as informative* for a given prediction.

For sparse MoE routing the analogous question is: *which input directions does a router’s gating function for expert e at layer ℓ treat as informative?* Adit’s framework gives a principled, non-empirical handle on that question through the AGOP of the gating network. The mechanistic chapter of the thesis estimates $M_{\ell,e}$ for selected experts and asks whether AGRS chooses experts whose top AGOP eigenvector aligns with concept-prototype directions, and whether STEERMOE does not. This is the specific intellectual bridge to the advisor’s existing toolkit and the reason this project is the right project to pursue under his supervision rather than under a generic ML-systems advisor.

4.5 Future plans

After this master’s I intend to pursue a research role in AI interpretability and behavioral evaluation of large language models, either at an industry research lab focused on alignment and post-training intervention methods, or via a PhD program. The thesis is sized to produce the publication artifact (a NeurIPS-tier method paper) and the cluster-scale experimental experience that admissions committees and research-team hiring rounds evaluate against. Even absent a PhD trajectory, the three deliverables — paper, open-source replication artifact, written monograph — stand on their own as evidence of the ability to design, execute, and write up an interpretability research program end-to-end.

5. Goals and Expected Deliverables

The thesis has three goals, each with a concrete deliverable, an assessable criterion of success, and a phase from the project’s [REVISION_PLAN.md](#).

- **Goal 1 — Method.** Define and implement AGRS as a drop-in replacement for STEERMoE on Mixtral 8x7B-Instruct and OLMoE 1B-7B-Instruct. The implementation is paired with unit tests that include a property test: *AGRS reduces to STEERMoE when the attention weights are uniform*. Success criterion: AGRS produces strictly more-concentrated risk-difference tables than STEERMoE on $\geq 4/5$ of the fear concepts in the pilot, measured by the top-20 share of $|\Delta|$ mass.
- **Goal 2 — Empirical evaluation.** A statistically powered, pre-registered comparison of AGRS against five alternatives — baseline, STEERMoE, ITI, ActAdd, and RepE difference-of-means — across four MoE checkpoints and four concept families. Sample size $n = 250$ cases per condition; five seeds. Primary metric is concept-transfer rate (CTR) from an LLM-as-judge cross-validated against a human-rater subset of 200 cases. Success criterion: AGRS lifts CTR by ≥ 0.10 over STEERMoE on Mixtral / fears with non-overlapping paired bootstrap 95% CIs, and produces a positive CTR shift in $\geq 3/4$ models and $\geq 3/4$ concept families.
- **Goal 3 — Mechanistic analysis with AGOP.** Estimate the AGOP of selected experts’ gating functions over the contrastive prompt pool and ask whether AGRS’s selections align with concept-distinctive directions. Success criterion: AGRS- selected experts have systematically higher cosine alignment between their top AGOP eigenvector and the concept-prototype embedding than STEERMoE-selected experts on Mixtral / fears, with the per-cell alignment-vs-CTR correlation across (model, concept) cells reaching $|r| \geq 0.5$.

Expected publication. A submitted NeurIPS or ICLR main-track paper, with the thesis monograph organized around the three goals above and a final chapter situating the empirical findings inside the AGOP feature-learning framework.

Expected artifact release. A public, anonymized `experiments/` tree under CC-BY containing every executed notebook, generation metadata, qualitative review, activation table, steering plan, judge transcript, and pre-registration document. The Apptainer image and pinned environment lock are part of the release.

6. Scientific Justification

Why this is interesting. Behavioral steering at inference time is one of the few corners of mechanistic interpretability where small, tractable interventions on frozen models produce measurable, causal behavioral effects. Most published steering methods operate on the residual stream of dense transformers. Sparse MoE models give us a structurally different intervention site — the router logit — whose causal effect is mediated through a discrete, inspectable selection over a small expert pool. Whether the readout-and-intervene paradigm transfers from dense to sparse architectures, and whether it does so with the *same* readout rules, is an open empirical question with direct implications for how alignment researchers deploy steering on open-weight MoE checkpoints (which now make up a large fraction of the open-weight frontier).

Why this is innovative. Three innovations distinguish the proposed thesis from existing work.

- *Method.* AGRS is, to my knowledge, the first attention-guided readout developed specifically for router-logit interventions on sparse MoE models. The reduction from AGRS to STEERMoE (under uniform attention) makes the contribution isolatable from confounding implementation

differences — the same intervention site, the same generation pipeline, the same experiment harness, the same evaluation. *Only the readout changes.*

- *Mechanistic framing.* I use Adit’s AGOP machinery to formalize the feature directions that selected experts respond to, then test whether AGRS’s selections align with those directions on the same concept-prompt pairs. This is a use of the feature-learning toolkit on a behavioral-intervention problem where it has not been applied; the result either validates the feature-learning lens on routing decisions or constrains where the framework applies.
- *Empirical rigor.* The evaluation is pre-registered with a Phase-3 decision gate ($\Delta\text{CTR} \geq +0.10$ over STEERMOE on the Mixtral / fears pilot) before the more expensive cross-model and cross-concept phases run. The full prompt set, judge transcripts, and seed sweeps are released under CC-BY at submission.

Why now. The empirical conditions for this thesis exist in a way they did not 18 months ago: open-weight Mixtral, OLMoE, and Qwen2-MoE checkpoints are all released and stable; LLM-as-judge evaluation with frontier models has become defensible enough to cross-check with smaller human-rater subsets; and reproducibility norms in the alignment community (pre-registration, full artifact release) have matured to the point that a master’s thesis can credibly ship a reviewer-grade reproducibility bundle. The window for a clean, narrow, reproducible MoE-steering contribution is open.

7. Literature Review

I group the literature into four threads. The thesis sits at the intersection of the third and fourth: it extends sparse-MoE intervention with a readout machinery taken from dense-transformer attention-guided steering, and it analyzes the result using feature-learning theory.

(i) Activation steering on dense transformers. The seminal example is *Inference-Time Intervention* (Li et al., NeurIPS 2023), which adds a learned direction to a small set of attention heads to reduce hallucination on TruthfulQA. *Representation Engineering* (Zou et al., 2023) generalizes the readout-and-edit problem to contrastive prompt pairs at the residual stream. *Activation Addition* (Turner et al., 2023) projects the contrastive signal across multiple residual layers via projected gradient descent. *Attention-Guided Steering* (Tan et al., 2024) refines *where* on the residual stream to read by weighting positions by attention mass; this is the direct upstream inspiration for the readout half of AGRS. None of these methods target sparse MoE routing.

(ii) Sparse Mixture-of-Experts language models. *Mixtral of Experts* (Jiang et al., 2024) and *OLMoE* (Muennighoff et al., 2024) released the open-weight checkpoints used in this thesis. Each sparse layer routes every token to two experts via a learned router. *Qwen2-MoE* (Yang et al., 2024) and *DeepSeek-MoE* (Dai et al., 2024) round out the cross-architecture set. The discrete routing decision is the structure behavioral steering exploits, and routing-pool size (8–64 experts) is one of the axes we expect AGRS’s gain to vary along.

(iii) Steering sparse MoE models. STEERMOE (2024) is the upstream work that first proposed biasing router logits to alter expert selection at decode time. Its readout uses uniform averaging over a target span. The diagnostic results in my preliminary draft show this readout produces a diffuse risk-difference signal on Mixtral and correspondingly small behavioral shifts under a question-only

test. The thesis builds directly on STEERMOE’s intervention site while replacing its readout; the same paper-rule parameters ($\varepsilon = 0.01$, $k_+ = 20$, $k_- = 0$) are kept as a sanity-check baseline.

(iv) Feature learning and AGOP. Radhakrishnan et al. (*Science*, 2024) introduce the *neural feature ansatz* and the AGOP as a mechanism through which neural networks acquire features. The CNN follow-up (arXiv:2309.00570) extends AGOP to convolutional architectures, and the Recursive Feature Machines paper (arXiv:2401.04553) gives RFM-based recovery guarantees for low-rank structure. *Quadratic Models for Understanding Neural Network Dynamics* (ICLR 2024) uses quadratic surrogates for training-dynamics analysis. The mechanistic chapter of the thesis uses AGOP to characterize which input-token directions a Mixtral expert’s gating function is sensitive to, and asks whether AGRS’s selections align with those directions.

8. Methodology

The project is mainly computational, as the IACS guidelines require. Every claim in the thesis is empirical and reproducible from a pinned codebase, a pinned environment, and a fixed set of model checkpoints. The methodology section spells out how that is enforced.

8.1 Models and compute

- *Models.* Mixtral 8x7B-Instruct-v0.1 (8-bit with CPU offload), OLMoE 1B-7B-0125-Instruct, Qwen2-MoE 14B-A2.7B-Instruct, and DeepSeek-V2-MoE-Lite-Chat if licensing and storage allow. Each is loaded once via Hugging Face Transformers under a recorded revision SHA, asserted by a helper on every load.
- *Compute.* Two NVIDIA A100 GPUs per generation job on the MIT ORCD `mit_normal_gpu` partition. Total budget: ~ 305 GPU-hours across all phases, sized for one graduate-student semester.
- *Software.* Python 3.11; PyTorch; Hugging Face Transformers; `bitsandbytes`; `accelerate`; `papermill` for headless notebook execution; `numpy`, `pandas`, `matplotlib` for analysis; Apptainer for cluster-portable images.

8.2 The AGRS readout, formally

For a contrastive prompt pair (p^+, p^-) with concept-bearing prefix span $\mathcal{C} \subset p^+$, let

$R_{\ell,t,e} \in \{0, 1\}$ — indicator that expert e is selected by top- k routing at layer ℓ , token position t ;

$A_{h,t \rightarrow t'}^{\ell^*}$ — attention weight from token t to token t' at reference layer ℓ^* , head h ;

$a_t = \max_h \sum_{t' \in \mathcal{C}} A_{h,t \rightarrow t'}^{\ell^*}$ — per-token attention mass directed toward the concept span at ℓ^* .

The AGRS attention-weighted activation rate on p^+ is

$$r_{\ell,e}^+ = \frac{\sum_{t \in \text{target}} a_t^+ \cdot R_{\ell,t,e}^+}{\sum_{t \in \text{target}} a_t^+},$$

with $r_{\ell,e}^-$ defined identically on p^- with $a^- \equiv 1$ (uniform, since p^- has no concept span). The per-cell risk difference is $\Delta_{\ell,e} = r_{\ell,e}^+ - r_{\ell,e}^-$. The decode-time intervention is identical to STEERMOE: add $+\alpha$ to selected positive experts’ router logits, $-\alpha$ to selected negative experts’. STEERMOE is recovered when $a_t^+ \equiv 1$, which makes the contribution isolable.

8.3 Computational pipeline

1. *Routing-trace and attention collection.* For each (p^+, p^-) , capture per-token, per-layer router logits and attention weights via Hugging Face forward hooks. Slice to the shared statement-body target span.
2. *Risk-difference table.* Compute $\Delta_{\ell,e}$ under both the AGRS and STEERMOE readouts on the same traces. The contrast (top- k share of $|\Delta|$ mass under each readout) is one of the headline diagnostics of the final paper.
3. *Sparse plan selection.* Globally select the top k_+ positive cells and top k_- negative cells, subject to $|\Delta_{\ell,e}| > \varepsilon$.
4. *Decode-time intervention.* Add $\pm\alpha$ to selected experts’ router logits at every forward pass during generation.
5. *Evaluation.* Score each generation with an LLM judge (GPT-4o by default, Claude as cross-check) on a 0–2 ordinal scale for thematic concept transfer. Cross-validate against a 200-case human-rater subset (Cohen’s κ , Spearman). Report paired bootstrap 95% CIs over seeds and paired t -tests on seed-collapsed deltas.

8.4 Mechanistic analysis with AGOP

For a selected expert e at layer ℓ , treat the expert’s gating function $g_{\ell,e}(x)$ as a real-valued map from token-position embeddings to the pre-routing logit, and estimate

$$M_{\ell,e} = \mathbb{E}_x \left[\nabla_x g_{\ell,e}(x) \nabla_x g_{\ell,e}(x)^\top \right]$$

over the contrastive prompt pool. Compare the top eigenvectors of $M_{\ell,e}$ against the concept-prototype embedding direction. The empirical question: *do the experts AGRS selects respond to feature directions that an AGOP-style analysis identifies as concept-relevant, or does AGRS find them incidentally through the activation-rate differential alone?*

8.5 Reproducibility infrastructure

The thesis adopts a single, strict reproducibility pattern that is implemented in the project’s [notebooks/](#) layer:

- **The notebook is the pipeline.** Each experimental step is a Jupyter notebook that declares its inputs as a `papermill parameters` cell, calls into a pure-function entrypoint in `src/.../pipelines/`, runs identically on a laptop (CPU, smoke mode) and on the MIT ORCD cluster (GPU, full mode) via a paired `slurm/notebooks/<name>.sbatch` wrapper, and archives the executed copy with full outputs into the run directory. No experimental result lands in `experiments/` that did not come from a `papermill`-executed notebook.

- **Determinism.** All generation calls use temperature 0, top- p 1.0, with explicit seeds threaded through every pipeline function.
- **Pinning.** A helper asserts the recorded Hugging Face revision SHA on every model load; the assertion fires before any compute.
- **Provenance.** Every run writes `experiments/<run_dir>/provenance.json` with the git SHA of the codebase, the cluster job id, the hostname, the wallclock, and the resolved papermill parameter values.
- **Containerization.** One Apptainer image, built from `containers/agrs.def`, is the only supported cluster runtime.
- **Pre-registration.** The AGRS hypothesis (primary metric, comparator, n , seeds, stop rule, Phase-3 decision gate) is registered on OSF before any AGRS-vs-STEERMOE comparison runs.
- **Public release.** An anonymized `experiments/` tree under CC-BY ships at submission so a reviewer can re-derive every figure offline.

8.6 Timeline and milestones

Phase	Months	Milestone
0. Reproducibility hardening	May–Jun 2026	<code>make paper</code> regenerates current draft on a clean checkout.
1. Evaluation harness	Jun–Jul 2026	Judge-vs-human Spearman ≥ 0.7 on 200-case subset.
2. AGRS implementation	Jul 2026	Unit tests pass; AGRS $\neq$ STEERMOE on same data.
3. Pilot (Mixtral / fears)	Aug 2026	Decision gate: $\Delta\text{CTR} \geq +0.10$.
4. Full method comparison	Sep–Oct 2026	Six-method table for Mixtral / fears.
5. Cross-model generalization	Nov 2026	Four MoE models, AGRS-vs-STEERMOE table.
6. Cross-concept generalization	Dec 2026	Four concept families on Mixtral.
7. Mechanistic AGOP analysis	Jan 2027	AGOP alignment scatter + per-cell correlation.
8. Statistics and writing	Feb–Mar 2027	Thesis monograph and conference submission.

8.7 Risks and contingency

Phase-3 decision gate. If AGRS does not clear the pre-registered $\Delta\text{CTR} \geq +0.10$ threshold on Mixtral / fears, the thesis pivots in one of two pre-declared directions: (a) the same attention-weighted readout is applied to a hidden-state intervention site, isolating the readout from the intervention site; or (b) the thesis becomes a focused diagnostic-and-AGOP study of why the routing signal is diffuse on Mixtral. Either pivot is registered in the OSF pre-registration so it cannot be backfit.

Cross-model port engineering. Each new MoE checkpoint requires a backend module analogous to the existing Mixtral one. The plan budgets one engineering week per model, with DeepSeek-MoE-Lite as the most fragile (storage, license); if it stalls, the thesis reports three models rather than four and the decision is documented in `checkpoint_card.md`.

Refusal-family release. The refusal-target concept family carries dual-use considerations; we engage with the institutional contact early in Phase 6 and ship a public-release go/no-go decision

before the family’s generations enter the paper.

AGOP estimation cost. AGOP requires gradients of the gating function through tens of thousands of tokens. Phase 7 budgets ~ 20 GPU-hours; if tight, AGOP is estimated at a single representative layer rather than all 32.

Companion materials in the project repository (`paper-submission` branch):

- Working conference draft: `paper/docs/paper/main.pdf`.
- Phased revision plan with code-to-build manifest: `paper/docs/REVISION_PLAN.md`.
- Projected final-paper shape and outcome scenarios: `paper/docs/EXPECTED_FINAL_PAPER.md`.
- Notebook-as-pipeline scaffolding: `notebooks/README.md` and `slurm/notebooks/README.md`.